

Exercise 6.1: Why Explainability? What are the advantages of explainable models in a safety-critical context? What are potential disadvantages or limitations of current explainability methods?

Advantages of Explainable Models

- Understand why the model made a decision
Example: See why the model predicted "Pedestrian Present".
- Helps find errors and bugs
Example: The model focuses on the sky instead of the pedestrian.
- Helps detect bias
Example: The model relies on lighting conditions rather than the actual object.
- Increases trust in the system
Example: Engineers can verify that the model is looking at the pedestrian.
- Supports safety certification
Example: Provides evidence that the model makes decisions based on relevant features.

Disadvantages / Limitations

- Different methods may give different explanations
Example: Grad-CAM and Saliency Map highlight different regions.
- Explanations can be misleading
Example: Heatmap highlights the road, but the model actually uses another feature.
- May not show the true reasoning of the model
Example: Explanation looks reasonable, but the real decision process is different.
- Does not guarantee safety or correctness
Example: Model highlights the pedestrian but still makes a wrong prediction.
- Complex models are still hard to fully explain
Example: Explaining every decision inside a deep ResNet is difficult.

Exercise 6.2: Local vs. Global Explainability Describe the difference between local and global explainability. Name one method for each kind and explain what question it answers.

Difference

- Local Explainability explains why the model made a prediction for a specific input.
Example: Why did this image get classified as "Pedestrian Present"?
- Global Explainability explains how the model behaves overall and what patterns it has learned.
Example: What features does the pedestrian model generally use for detection?

One Method for Each

- Local Explainability: Grad-CAM
Question answered: "Which part of this image influenced the prediction?"
- Global Explainability: Filter Visualization
Question answered: "What patterns or features has the model learned overall?"

Exercise 6.3: Saliency vs. Occlusion Briefly describe the saliency method and the occlusion method. What is one advantage and one disadvantage of each compared to the other?

Saliency Method

- Measures how sensitive the prediction is to changes in input pixels using gradients.
- Highlights pixels that most influence the prediction.

Advantage: Fast because it only requires gradient computation.

Disadvantage: The resulting heatmaps can be noisy and difficult to interpret.

Occlusion Method

- Repeatedly covers parts of the image and observes how the prediction changes.
- Regions causing large prediction changes are considered important.

Advantage: Easy to understand and directly relates importance to prediction changes.

Disadvantage: Computationally expensive because the model must evaluate many occluded versions of the image.

Exercise 6.4: Chain-of-Thought Fidelity Modern LLMs often produce a thinking trace (chain of thought) alongside their answer.

1. What does it mean for a thinking trace to be faithful? Why is faithfulness hard to verify?

- A faithful thinking trace reflects the actual reasoning used by the model to reach its answer.
- Faithfulness is hard to verify because we cannot directly observe the model's internal decision-making process.
- The model may produce a convincing explanation that is different from its true reasoning.

Example: A student says "traffic jam" caused them to be late, but the real reason was oversleeping.

2. What is simulatability as a measure of thinking-trace quality? Give an example scenario where simulatability is satisfied but the trace is still unfaithful.

- Simulatability measures whether a person can predict the model's answer by reading the thinking trace.
- A trace can be simulatable but still unfaithful if the explanation is not the real reason behind the answer.

Example: The model guesses the correct answer first and then generates a logical-looking explanation afterwards. A reader can predict the answer from the explanation, but the explanation was not actually used to reach it.

3. What does counterfactual simulatability add? Why is it a stricter test?

- Counterfactual simulatability checks whether the reasoning still works when the question is slightly changed.
- It is stricter because it tests whether the explanation genuinely captures the reasoning process rather than just explaining one specific answer.

Example: If the reasoning says "add the two numbers," it should still produce the correct answer when the numbers are changed.

4. Why does an unfaithful thinking trace pose a safety risk? Give one concrete example.

- An unfaithful trace can make users trust a model for the wrong reasons.
- Engineers may think the model is reasoning correctly when it is actually relying on irrelevant factors.

Example: A medical AI explains that it diagnosed a disease using symptoms, but it actually relied mostly on patient age. Doctors may trust the diagnosis even though the reasoning is incorrect.

Exercise 6.5: Applying an Explainability Method Choose one local explainability method from the lecture: occlusion, CAM, Grad-CAM, or saliency maps. Apply it to each of your three CARLA models.

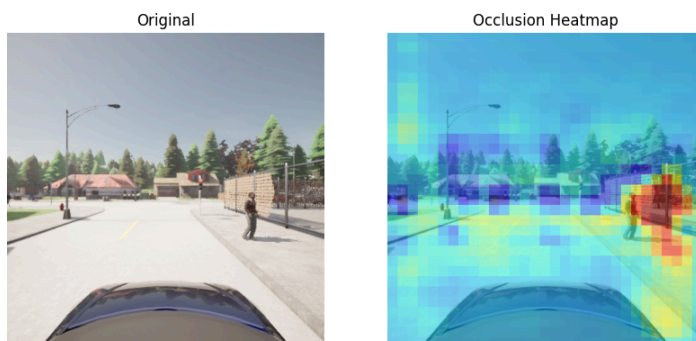
1. Briefly describe how your chosen method works and why you chose it.

Occlusion works by covering different regions of an image and observing how the prediction changes. If covering a region significantly decreases the prediction score, that region is considered important. I chose this method because it is simple, intuitive, and easy to interpret.

2. Select five correctly classified images - at least one per model. Visualize the explanation (heatmap or sensitivity map) overlaid on the original image

3. Do the highlighted regions correspond to the relevant objects (e.g., does the map highlight the pedestrian when the model predicts "pedestrian present")?

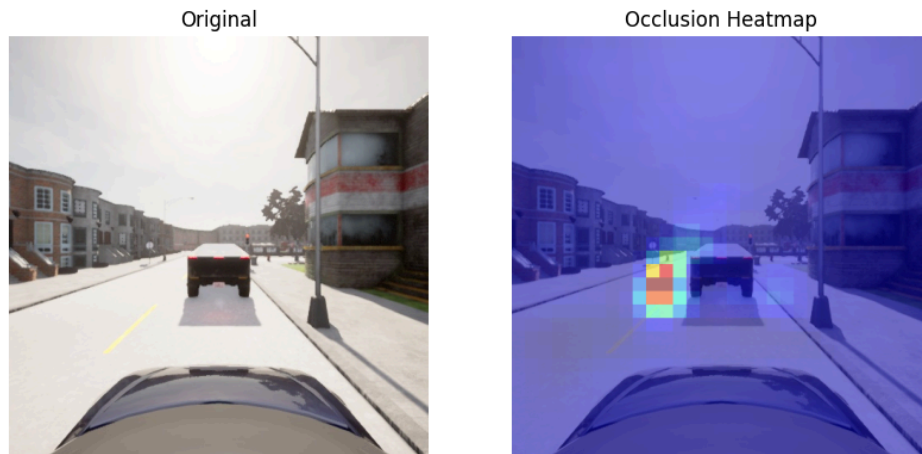
Pedestrian Model (Image 1):



The heatmap highlights the region around the **pedestrian on the right side of the road**.

The model appears to focus on the pedestrian when predicting "pedestrian present", suggesting that it has learned meaningful pedestrian-related features rather than relying solely on background information.

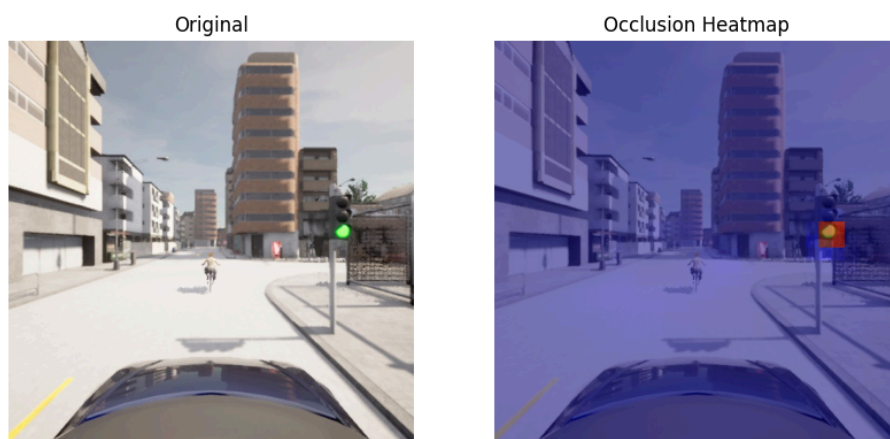
Vehicle Model (Image 2):



The strongest activation is located on the **vehicle in front of the ego vehicle**.

The model correctly focuses on the vehicle itself, indicating that the prediction is based on vehicle-related features rather than road or building structures.

Traffic Light Model (Image 3):

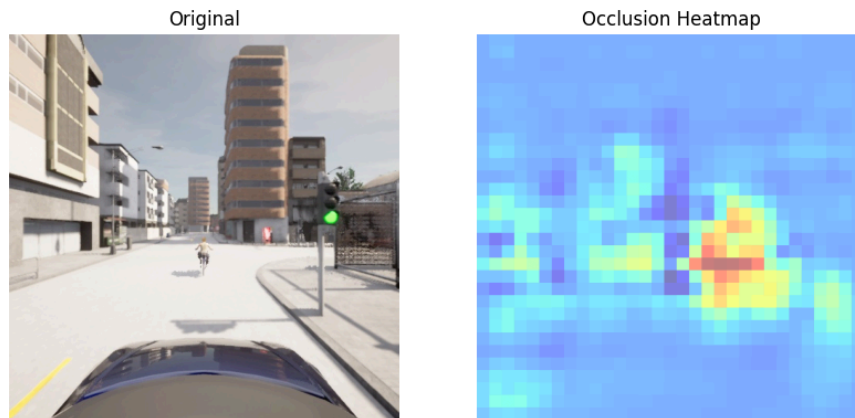


The heatmap clearly highlights the **traffic light on the right side**, especially the illuminated green light.

The model successfully identifies and focuses on the traffic light, suggesting that it learned features associated with traffic light structures and signal colors.

4. Select three misclassified images and apply the same method. Do the explanations give any insight into why the model failed?

Misclassified_ped Image 1:

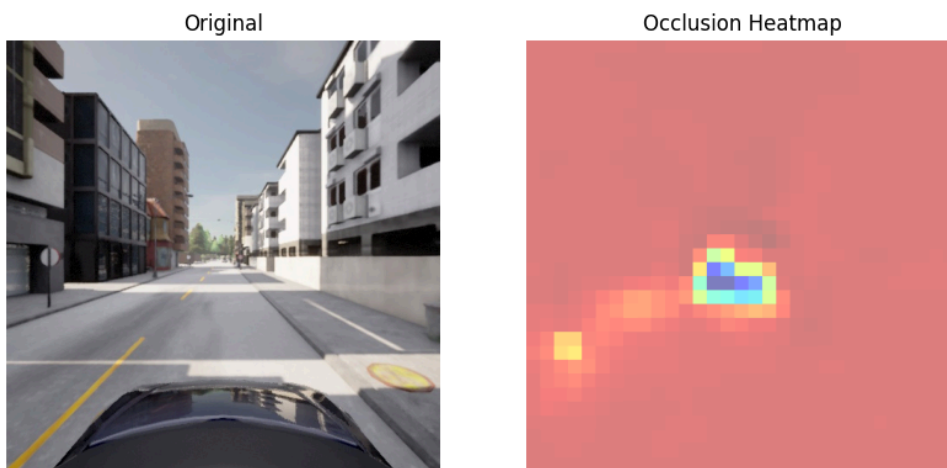


The heatmap focuses mainly on the traffic light and surrounding background, while the pedestrian in the scene receives little attention.

Insight:

The model appears to rely more on nearby objects and scene context than on the pedestrian itself. Because the pedestrian is relatively small and distant, the model may have failed to recognize it as the most important object.

Misclassified-Veh Image 2:

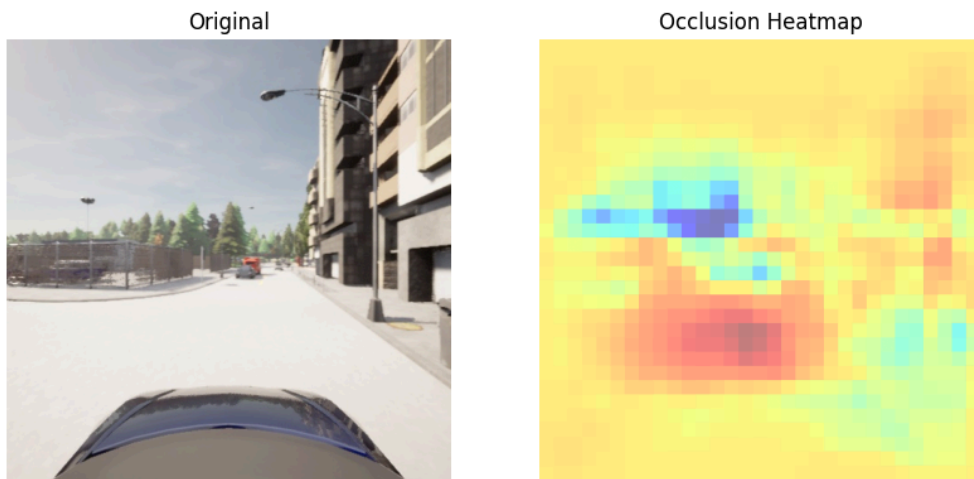


The strongest activation is concentrated on a small region in the center of the image rather than on a meaningful road object.

Insight:

The model seems to focus on irrelevant visual patterns instead of important scene features. This suggests that the prediction may have been influenced by background textures or image artifacts rather than the intended target object.

Misclassified-TL Image 3:



The heatmap is spread across large background regions, including buildings and road areas, with no clear focus on the relevant object.

Insight:

The model does not appear to have identified a specific object of interest. Instead, it relies on surrounding environmental features, which may explain the incorrect prediction. This indicates poor localization of the target object.

Exercise 6.6: Explainability as a Diagnostic Tool 1. Suppose your explanation method reveals that your model predicts “pedestrian present” based primarily on a region of the sky rather than the pedestrian itself. What would this imply about the model’s generalization? What could cause this behavior?

- If the model predicts "pedestrian present" mainly based on the sky, it suggests poor generalization because the model is not learning the actual pedestrian features.
- The model may perform well on training data but fail when weather, lighting, or background conditions change.
- This behavior indicates that the model relies on spurious correlations rather than meaningful pedestrian characteristics.

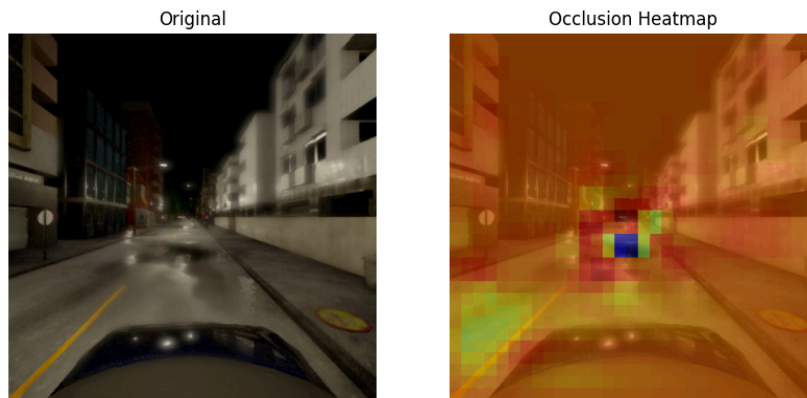
Possible causes:

- Dataset bias in the training data.
- Lack of diverse training examples.
- Correlation between background features and labels.
- Overfitting to irrelevant scene characteristics.

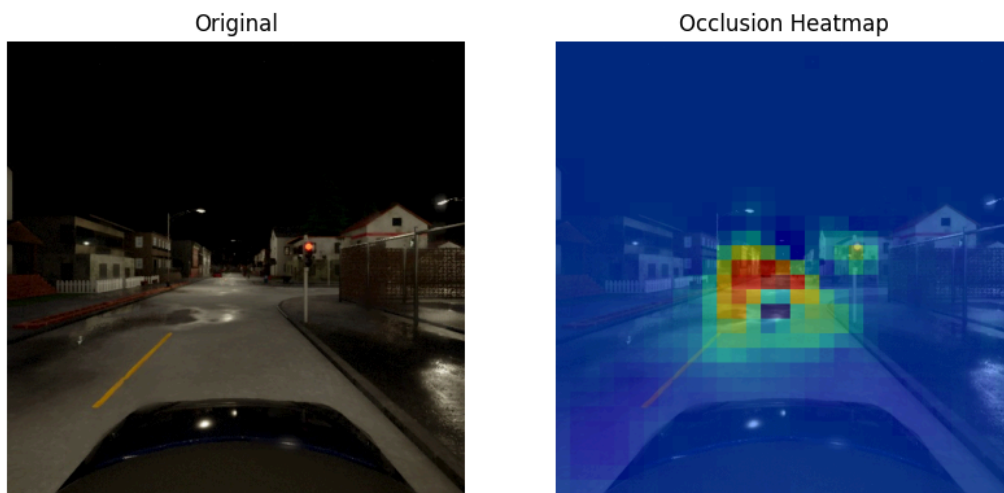
Example: If most pedestrian images in the training set were captured under a blue sky, the model may incorrectly associate the sky with the presence of pedestrians instead of learning human shapes

2. Now test your models on images from different conditions than the ones they were trained on - for example a different town layout, nighttime, or foggy weather. Apply the same explanation method as before.

Correct prediction:



Wrong prediction:



(a) Do the highlighted regions still correspond to the relevant objects?

- Yes, the heatmaps still focus on important regions of the scene.
- The model is not looking at completely random areas.
- However, the focus is less clear compared to the original daytime images.

Example: In daytime images, the heatmap clearly highlighted the pedestrian or traffic light. In nighttime images, the highlighted area is more spread out

(b) Do you observe evidence that the model relied on background or spurious features?

- Yes, some evidence can be observed.
- The heatmaps also highlight road reflections and bright lighting areas.
- This suggests the model may be using lighting conditions in addition to the target object.
- Such features may not generalize well to different environments.

Example: The model pays attention to bright reflections on the road instead of focusing only on the object.

(c) How do the model's accuracy and explanation quality change across conditions?

- Accuracy decreased under nighttime conditions.
- The heatmaps became less focused on specific objects.
- More attention was given to environmental features such as lighting and reflections.
- This indicates weaker generalization under distribution shift.

Example: In daytime images, the model focused mainly on the object. In nighttime images, the focus became more spread across the scene.